BERT: Bidirectional Encoder Representations from Transformers

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What is BERT?

- Developed by Google (2018)
- Pre-trained NLP model
- Based on Transformer architecture
- Understands context from both directions (bidirectional)

Why BERT?

1

Traditional models:

- Read text left → right OR right → left

2

BERT:

- Reads both directions at the same time
- Better understanding of meaning

Example:
“bank” (river vs money) → BERT understands from context

01 BERT Architecture

- Based on Transformer Encoder

Key components:

- Token Embeddings
- Segment Embeddings
- Position Embeddings

Uses:

- Self-Attention mechanism

02 Transformer

- Input sentence → tokens
- Passed through multiple encoder layers
- Each layer:
  - Attention
  - Feed-forward network

03 Pre-training Tasks

- Masked Language Model (MLM)Predict missing words
- Next Sentence Prediction (NSP)Understand sentence relationships

04

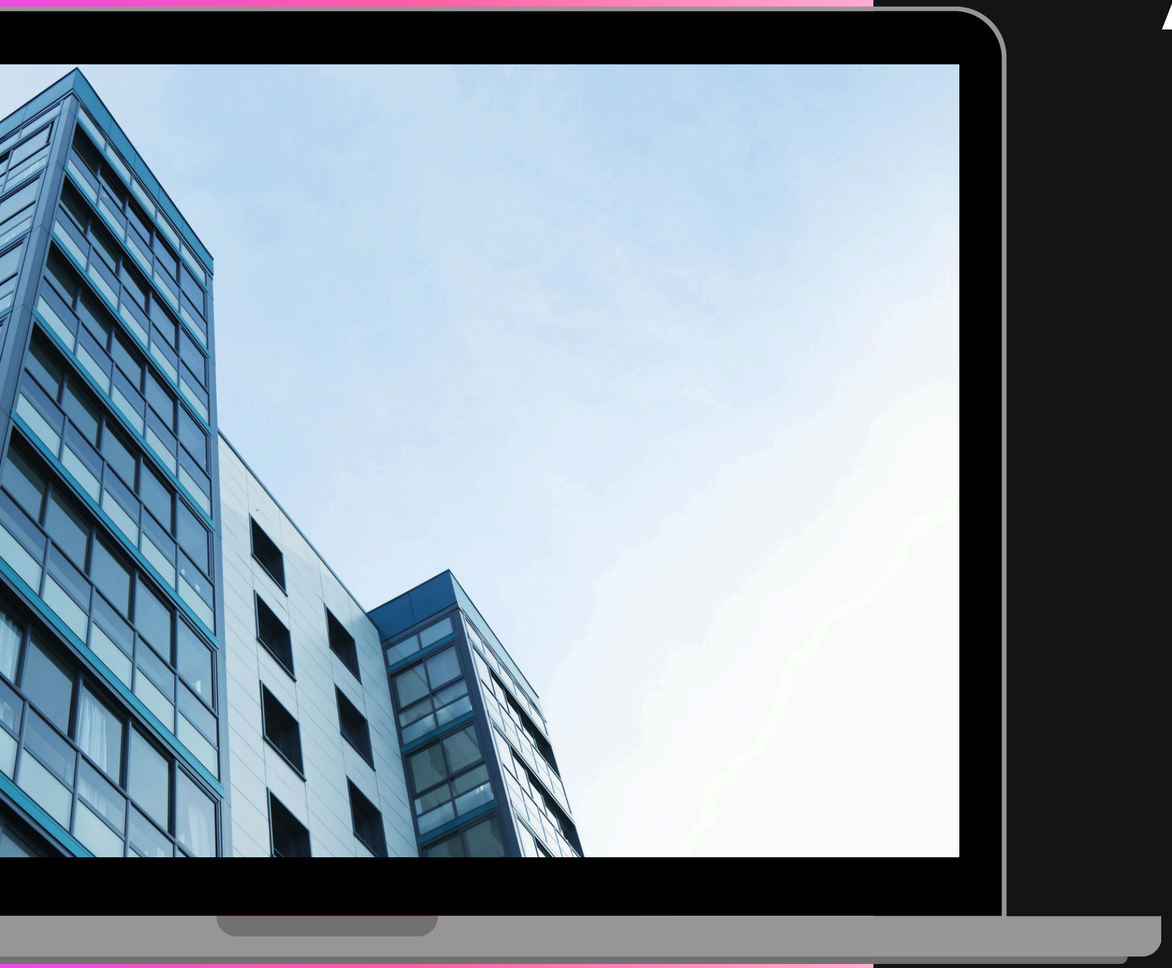
Training Libraries

Popular tools:

- Hugging Face Transformers
- PyTorch
- TensorFlow

Applications

- Sentiment Analysis
- Question Answering
- Chatbots
- Text Classification



Advantages

- Context-aware
- Pre-trained → saves time
- State-of-the-art performance

Limitations

- Heavy model (slow)
- Needs GPU
- Large memory usage

Conclusion

- BERT revolutionized NLP
- Widely used in modern AI systems